Deliver

Operate, learn and rehearse rollback

Observability, human review and rollback Session 5 of 5 · 3 hours · source-aligned workshop

What the final release file must explain

- What changed and which artifacts produced the answer.
- Which measured gates passed, failed or remain unknown.
- Who can approve release and who can restore service.

The last three-hour workshop

25 min operating loop + 20 min trace walkthrough

90 min labs, including a 10 min break

25 min incident review + 20 min final debrief

Showcase / a complaint becomes a test

SIGNAL

A staff member flags an unhelpful answer.

HUMAN REVIEW

A reviewer checks sources and accepts the signal.

REGRESSION ASSET

A separate curated answer becomes a golden case.

Run the reviewed-feedback demonstration

```
python -m training.showcase --session 5
```

No token: 401; pending: 409; curated export: 200.

Validate JSONL, keep the review, rehearse recovery.

Orchestration and Application Logic

- Educative lesson 14

- Concept → worked example → NexaAssist application

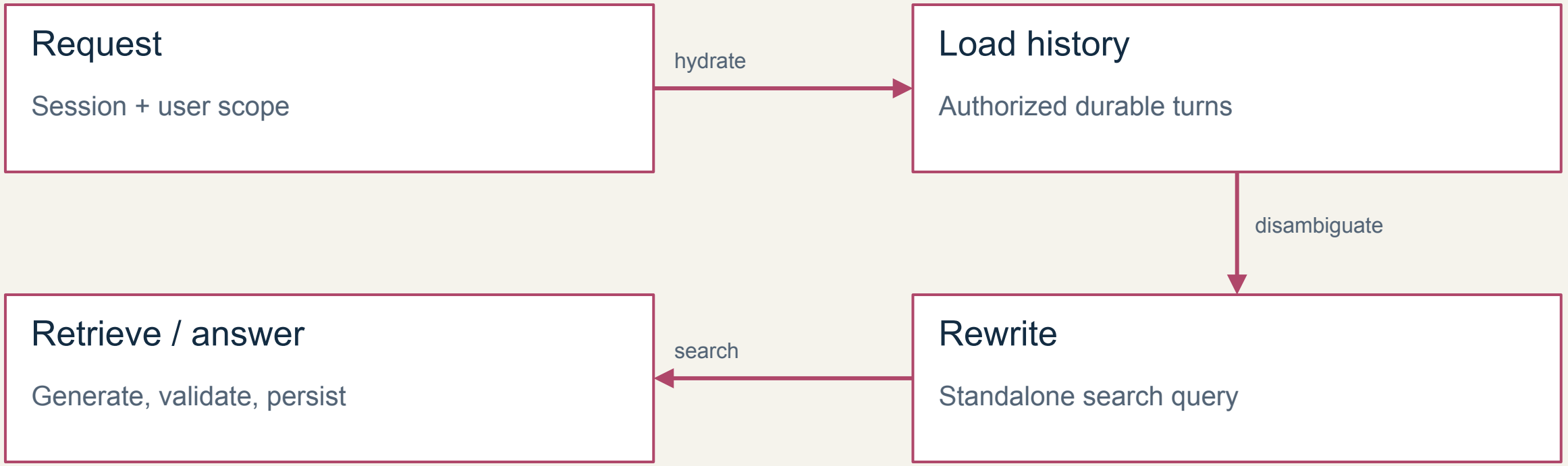
Follow-up questions change retrieval requirements

A query such as “Does that fee apply earlier?” lacks scope alone.

Conversation history can resolve the subject and date.

Memory adds privacy, latency and correctness obligations.

Keep API workers stateless; store conversation state



A sliding window is a budget approximation

Load a bounded recent history instead of all turns.

Message count is simple but ignores varying token length.

Reserve tokens for instructions, evidence and output.

Rewriting must preserve the user’s meaning

- Resolve references using only authorized recent history.
- Keep original question and rewritten search query separate.
- Validate output; time out or fall back when rewriting fails.

New state needs multi-turn evaluation

- Capture history, current input and expected source.
- Test incorrect rewrites, missing history and cross-user access.
- Recheck retrieval and answer quality before rollout.

Practical / model the memory contract offline

```
Run: python -m training.session5.memory_design
```

Inspect a bounded fixture and a failed-rewrite fallback.

Explain why the simulation is not a durable session service.

PAIR LAB / Predict → run → inspect → explain / uv run --no-sync

Quality, Human-in-the-Loop, and Governance Systems

Educative lesson 15

Concept → worked example → NexaAssist application

Use two kinds of observation

System health: latency, failures, tokens and priced cost.

Answer quality: grounding, relevance and correct scope.

A healthy system can still produce a harmful answer.

Trace one answer to its artifacts

- Prompt ID, version and content fingerprint.
- Chat/embedding model and retrieval configuration.
- Evidence references and optional Datadog span ID.

Sampled judging is an explicit choice

```
def should_sample(answer: AnswerResponse, settings: Settings) -> bool:
    """Only answered responses carry claims to check; abstentions are skipped."""
    return answer.status == "answered" and random.random() < settings.live_eval_sample_rate


async def score_sampled_answer(
    request: RetrieveRequest, answer: AnswerResponse, settings: Settings
) -> None:
    """Background reference-free judging of live traffic; failures are logged, never raised.
```

The sample rate is configurable; zero disables sampled judging.

Know what sampled faithfulness measures

> Sampling answered requests adds model cost.

> The current judge uses cited quotes as its context.

> This is a grounding proxy with limited observation.

Quality is maintained in three layers

> Automated gates catch known regressions.

> Human review assesses context and newly observed failures.

> Governance controls changes to meaning and risk.

The review queue needs the exact interaction

- Link the input, evidence references, output and versions.
- Prioritize explicit complaints and representative samples.
- Review the source before accepting the correction.

Human feedback is a signal

> A thumbs-down may mean wrong scope or poor wording.

> Review source, context and answer before accepting it.

> Never turn raw feedback directly into a golden answer.

The human review loop has owners

1. Signal → triage → source-supported correction
2. Review → proposed golden case → regression replay
3. Approved fix → measured release → observation

Practical 2 / inspect feedback behavior

```
Run feedback, sampling and comparison tests.
```

Follow reviewer authentication and review transitions.

Explain why accepted feedback still needs a verified reference.

PAIR LAB / Predict → run → inspect → explain / uv run --no-sync

Export a reviewed regression case

```
Run: python -m training.session5.feedback_loop
```

Accepted review + curated answer → golden case.

Keep the review bundle beside the validated JSONL.

PAIR LAB / Predict → run → inspect → explain / uv run --no-sync

Governance follows the change’s meaning

> Engineering reviews code and operational behavior.

> A domain expert reviews policy meaning and golden answers.

> The designated owner approves real data use and risk.

Watch both semantic and operational drift

> Topic shifts can expose gaps in the golden set.

> Policy updates can invalidate otherwise stable answers.

> Judge changes can move scores without application changes.

Practical 1 / run an incident safely

```
Run: python -m training.session5.incident
```

No traffic → every required SLO is not_measured.

Scenario failures: p95 latency and hallucination proxy.

PAIR LAB / Predict → run → inspect → explain / uv run --no-sync

A regression needs a comparable baseline

> Scenario faithfulness changes from 0.95 to 0.70.

> Delta −0.25 exceeds the placeholder tolerance.

> Investigate cases and configuration before interpreting it.

Exercise / remove an observation

```
Set sampled_count=0 and the proxy to None.
```

Predict not_measured rather than pass.

Explain what evidence is needed before clearing an incident.

PAIR LAB / Predict → run → inspect → explain / uv run --no-sync

Practical 3 / rehearse rollback

1. Hold candidate → preserve evidence → restore known-good set
2. Replay failing and golden cases → measure representative traffic
3. Document owner decision and follow-up regression case

A new index name is not a data snapshot

- Ingestion mutates the configured chunk collection.
- Parent sections must also match the candidate corpus.
- Blue/green requires isolated compatible stores and a tested switch.

What belongs in the release file

> Prompt/model/retrieval identities and dataset hash.

> Offline quality report and representative SLO evidence.

> Approver, rollback owner, known limits and incident links.

Future Directions in LLMOps

- Educative lesson 16

- Concept → worked example → NexaAssist application

Agents add actions and new failure modes

> Tool access needs permissions and an allowlist.

> Loops need iteration, time and cost bounds.

> Side effects need isolation, audit and recovery.

Bound an agent before granting actions

- Limit steps, time, token spend and tool permissions.
- Isolate side effects and preserve an audit trail.
- Evaluate tool choice, arguments and recovery behavior.

Multimodal inputs expand the ingestion boundary

- OCR or vision extraction introduces interpretation errors.
- Images and audio can carry identifiers or instructions.
- Queue expensive processing and evaluate the extracted evidence.

Model distillation introduces a training lifecycle

- Curate teacher traces into reviewed training data.
- Evaluate a smaller student on quality and safety cases.
- Version the dataset, model and rollback decision.

Edge execution shifts the operational trade-off

- Local inference can reduce network disclosure and round trips.
- Device differences affect performance and version consistency.
- Telemetry, updates and recovery need a new design.

The lifecycle survives a change in capabilities

- Define the new permission and data boundaries.
- Evaluate before deployment and version the artifacts.
- Observe failures and close the reviewed feedback loop.

Final gate / defend the decision

- Submit manifest, retrieval experiment and golden review.
- Attach quality/SLO reports and an incident write-up.
- State release, hold or unmeasured with supporting evidence.

Operate the loop, not just the endpoint

- Version the artifacts that change answers.
- Measure evidence, meaning and operations separately.
- Turn reviewed failures into durable regression cases.